Sensors and State Estimation

UAVs

MEAer - Autumn Semestre – 2022/2023

Rita Cunha / J.R.Azinheira

Sensors – Beard & McLain

Sensors for MAVs

- The following types of sensors are commonly used for guidance and control of MAVs
  - accelerometers
  - rate gyros
  - pressure sensors
  - magnetometers (digital compasses)
  - GPS

AR-drone:
+ sonar
+ vision (optical flow)

(no GPS)

Beard & McLain, "Small Unmanned Aircraft," *Princeton University Press*, 2012, Chapter 7, Slide 2

Sensors : AR.Drone

Sonar :

- Ultrasonic sensors are used to estimate height and vertical displacements of the UAV.
- They can also be used to determine the depth of the scene observed by the vertical camera.
- The ultrasonic sensors have a 40 kHz resonance frequency and can measure distances as large as 6 m at a 25 Hz rate.

Sensors : AR.Drone

Vision :

- The speed estimation from the images provided by the vertical camera (pointing to the ground in hovering flight) is achieved using two alternative algorithms:

1. The first algorithm is a multi-resolution scheme and computes the optical flow over the whole picture range.

2. The second algorithm, usually named “corner tracking”, estimates the displacement of several points of interest (trackers).

Sensors and State Estimation

(additional notes)

UAVs

MEAer - Autumn Semestre – 2022/2023

-revisiting accelerometers, rate gyros, magentometers
-Kalman Filter as na optimal observer
-KF for attitude estimation: rol measurement exemple
-Complementary filters

Rita Cunha / J.R.Azinheira

Accelerometers – basic principle

- h Inertial height of casing
- $h_1 = h + \delta$ Inertial height of proof mass
- δ Relative displacement

$$m\ddot{h}_1 = -k\,(h_1 - h) - \beta(\dot{h}_1 - \dot{h}) - mg$$

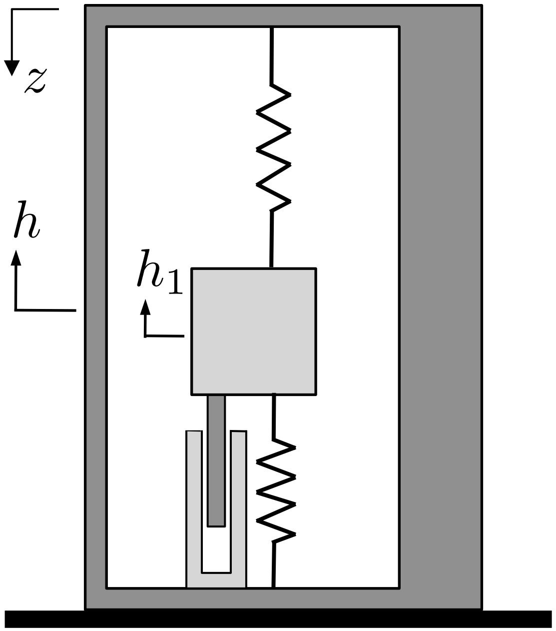
  – change of variables

$$\delta = h_1 - h$$

$$z = -h$$

$$m(\ddot{h} + \ddot{\delta}) = -k\delta - \beta\dot{\delta} - mg$$

$$m\ddot{\delta} + \beta\dot{\delta} + k\delta = m(\ddot{z} - g)$$

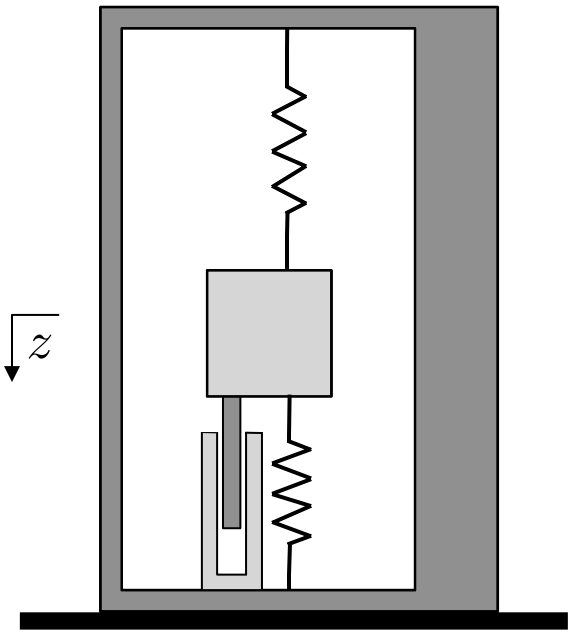


How a smartphone knows up from down (accelerometer)
https://www.youtube.com/watch?v=KZVgKu6v808

Accelerometers – basic principle

$$m\ddot{\delta} + \beta\dot{\delta} + k\delta = m\ddot{z} - mg$$

$$D(s) = \frac{1}{s^2 + \frac{\beta}{m}s + \frac{k}{m}}(A_z(s) - g\frac{1}{s})$$



- Within the low frequency range

$$\delta \approx \frac{m}{k}(\ddot{z} - g)$$

- 3-axis accelerometers attached to the vehicle (body-fixed frame {B})

$$\delta \propto R^T(\ddot{p} - ge_3) = (S(\omega)v + \dot{v}) - gR^T e_3$$

Accelerometers – basic principle

- 3-axis accelerometers attached to the vehicle (body-fixed frame {B})

$$\delta \propto R^T(\ddot{p} - ge_3) = (S(\omega)v + \dot{v}) - gR^T e_3$$

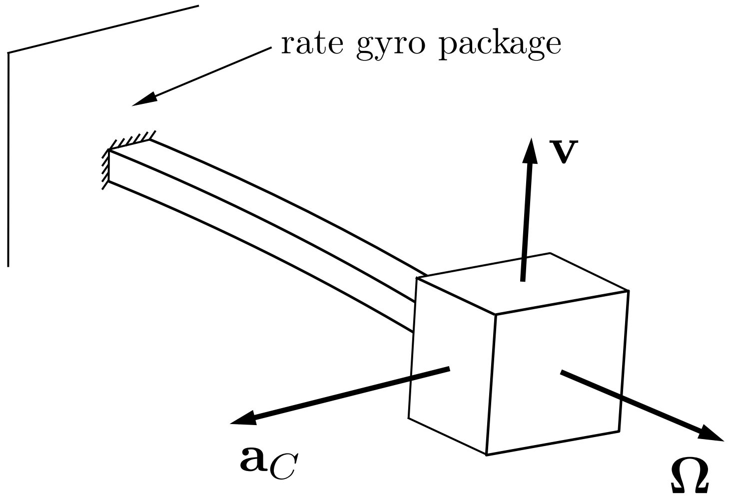
- Near hover condition

$$\delta \propto -gR^T e_3 = -g \begin{bmatrix} -\sin\theta \\ \cos\theta \sin\phi \\ \cos\theta\cos\phi \end{bmatrix}$$

Accelerometers are typically used for **attitude estimation** in combination with other sensors.

Rate gyros – basic principle

- Proof mass with position p and actuated vibration velocity v
- Angular velocity to be measured from resulting Coriolis acceleration



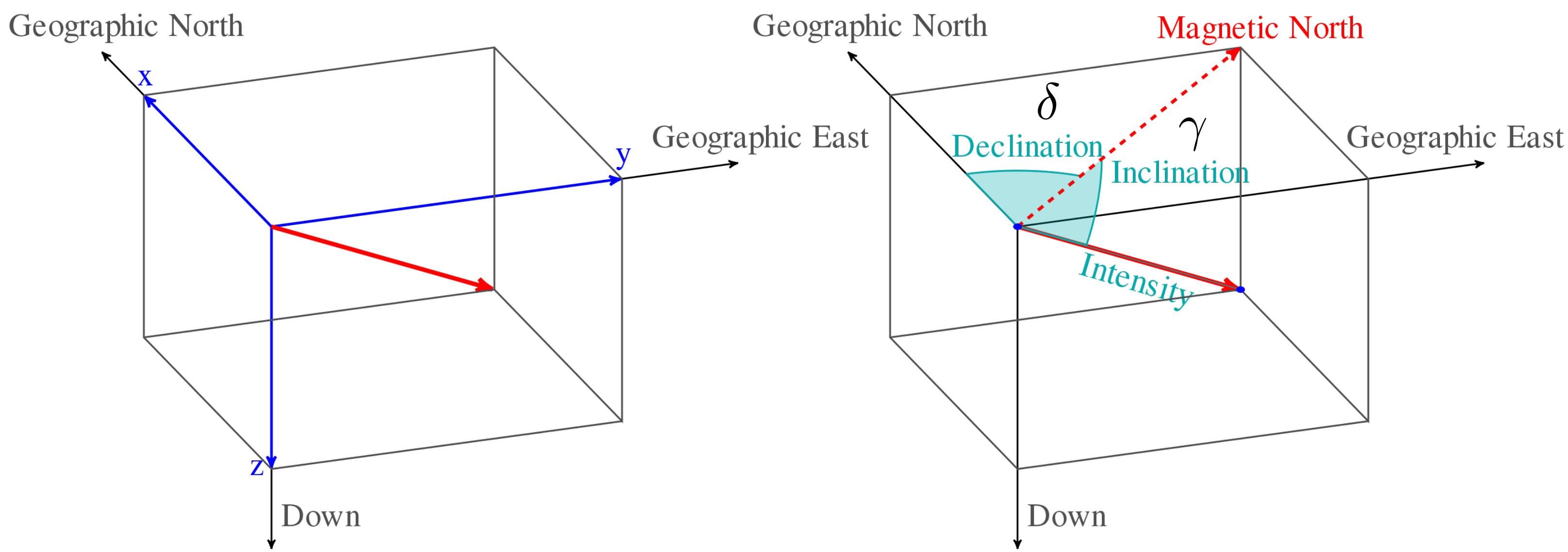
$$
\begin{aligned}
p_1 &= R^T p \\
\dot{p}_1 &= -S(\omega) R^T p + R^T \dot{p} \\
\ddot{p}_1 &= -S(\dot{\omega}) R^T p + S(\omega)^2 R^T p - 2S(\omega) R^T \dot{p} + R^T \ddot{p}
\end{aligned}
$$

$$
\omega = \begin{bmatrix} 0 \\ 0 \\ \Omega \end{bmatrix} \quad \dot{p}_1 = \begin{bmatrix} v_x \\ 0 \\ 0 \end{bmatrix}
$$

$$
\ddot{p}_1 = -\begin{bmatrix} 0 \\ 2\Omega v_x \\ 0 \end{bmatrix} + R^T \ddot{p}
$$

Proportional to the angular velocity that we would like to measure

Magnetometers



- **Available data:** $${}^{I}m_0 = \begin{bmatrix} \cos\delta\cos\gamma \\ \sin\delta\cos\gamma \\ \sin\gamma \end{bmatrix}, \quad {}^{B}m_0 = {}^{B}_{I}R\,{}^{I}m_0, \quad \phi, \quad \theta$$

  Depends on location

  Provided by 3-axis magnetometer

  Estimated using other sensors

- **Observed unknown:** ψ

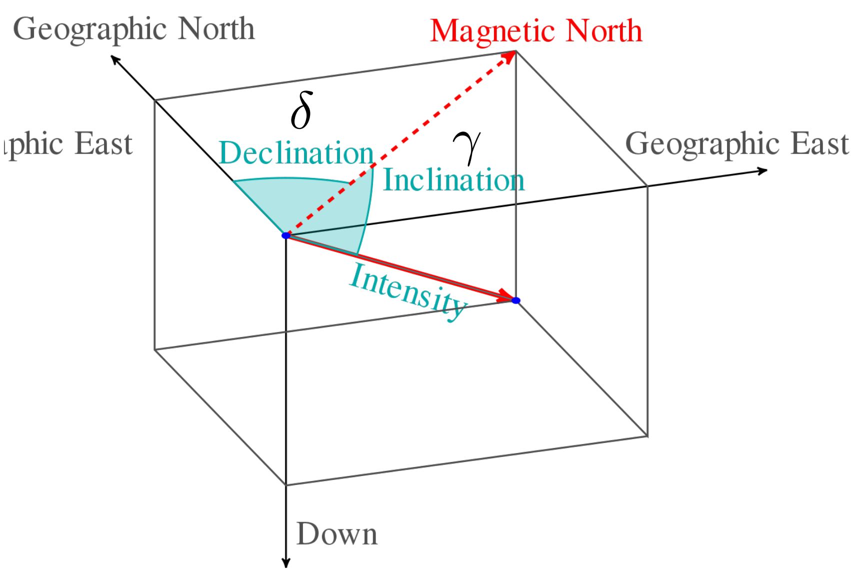
Magnetometers

- Available data: $^{I}m_0 = \begin{bmatrix} \cos\delta\cos\gamma \\ \sin\delta\cos\gamma \\ \sin\gamma \end{bmatrix}, \quad {}^{B}m_0 = {}^{B}_{I}R\,{}^{I}m_0, \quad \phi, \quad \theta$
- Observed unknown: ψ

$$^{B}m_0 = {}^{B}_{I}R\,{}^{I}m_0 = R_x(-\phi)R_y(-\theta)R_z(-\psi)\,{}^{I}m_0$$

$$R_y(\theta)R_x(\phi)\,{}^{B}m_0 = R_z(-\psi)\,{}^{I}m_0$$

$$^{B_1}m_0 = \begin{bmatrix} \cos(-\psi+\delta)\cos\gamma \\ \sin(-\psi+\delta)\cos\gamma \\ \sin\gamma \end{bmatrix}$$



$$(\psi - \delta) = \text{atan2}(-{}^{B_1}m_y, {}^{B_1}m_x)$$

Kalman Filter

- State observer/estimator for a stochastic system :

$$\dot{x} = Ax + Bu + w$$

$$y = Cy + n$$

  - the disturbance *w* and the measurement noise *n* are zero-mean white noise stochastic processes with covariance matrices

$$E[w(t)w(\tau)^T] = \delta(t-\tau)W$$

$$E[n(t)n(\tau)^T] = \delta(t-\tau)N$$

- Kalman filter design provides state estimate that minimizes

$$J = \lim_{t \to +\infty} E[\tilde{x}(t)^T \tilde{x}(t)]$$

  - Estimation error $\tilde{x}(t) = x(t) - \hat{x}(t)$

Kalman Filter

- Kalman filter design provides state estimate that minimizes

$$J = \lim_{t \to +\infty} E[\tilde{x}(t)^T \tilde{x}(t)] = \lim_{t \to +\infty} tr[\Sigma(t)]$$

  – Error covariance $\Sigma(t) = E[\tilde{x}(t)\tilde{x}(t)^T]$

- Solution takes the form of a state observer

$$\dot{\hat{x}} = A\hat{x} + Bu - L(y - C\hat{x})$$

  – The gain L is given by $L = \Sigma_\infty C^T N^{-1}$

where Σ_∞ is the steady-state error covariance, given by the solution to

$$(A - LC)\Sigma_\infty + \Sigma_\infty(A^T - C^T L^T) + W + LNL^T = 0$$

Duality between LQR and Kalman Filter

LQR design

$$\dot{x} = Ax + Bu$$

$$u = -Kx$$

$$\dot{x} = (A - BK)x$$

$$J = \int_0^{+\infty} (x^T Q x + u^T R u)dt$$

$$Q = C^T C \geq 0, \ \ R > 0$$

(A, B) Stabilizable

(A, C) Detectable

$$K = R^{-1} B^T P$$

$$PA + A^T P + Q - PBR^{-1}B^T P = 0$$

Kalman filter design

$$\dot{x} = Ax + Bu + \bar{B}w$$

$$y = Cx + n$$

$$\dot{\hat{x}} = A\hat{x} + Bu + L(y - C\hat{x})$$

$$\dot{\tilde{x}} = (A - LC)\tilde{x} + \bar{B}w + Ln$$

$$J = \lim_{t \to \infty} E[\|\tilde{x}(t)\|^2]$$

$$w \sim \mathcal{N}(0, (t - \tau)W)$$
$$n \sim \mathcal{N}(0, (t - \tau)N),$$

(A, C) Detectable

$(A, \bar{B})$ Controllable ?

$$L = \Sigma C^T N^{-1}$$

$$\Sigma A^T + A\Sigma + W - \Sigma C^T N^{-1} C\Sigma = 0$$

Kalman filter – exemple: roll angle measurement

- Roll angle estimation for an aircraft (fixed-wing)
  - Assumption: coordinated turn, no wind, no sideslip

$$F_{lift} \cos\phi = mg$$

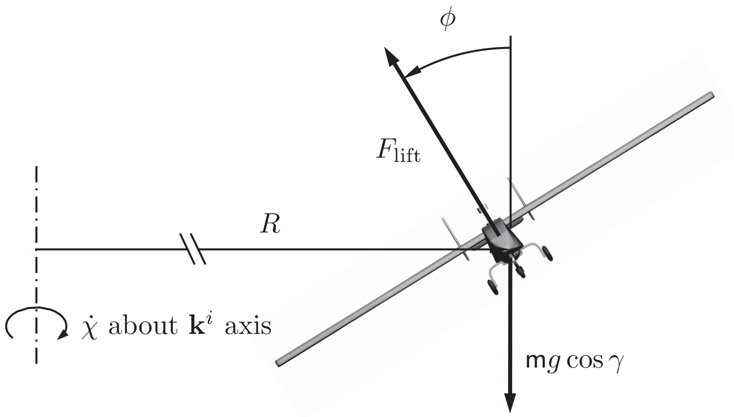
$$F_{lift} \sin\phi = mV^2/R = mV\dot{\psi}$$

$$\downarrow$$

$$\tan\phi = \frac{V\dot{\psi}}{g}$$

$$\downarrow$$

$$\dot{\psi} = \frac{g}{V}\tan\phi$$



  - Sensors: rate gyros
    (may not be enough …)

Kalman filter – exemple: roll angle measurement

- Roll angle estimation for an aircraft
  - Assumption: coordinated turn, no wind, no sideslip

$$\dot{\psi} = \frac{g}{V} \tan \phi$$

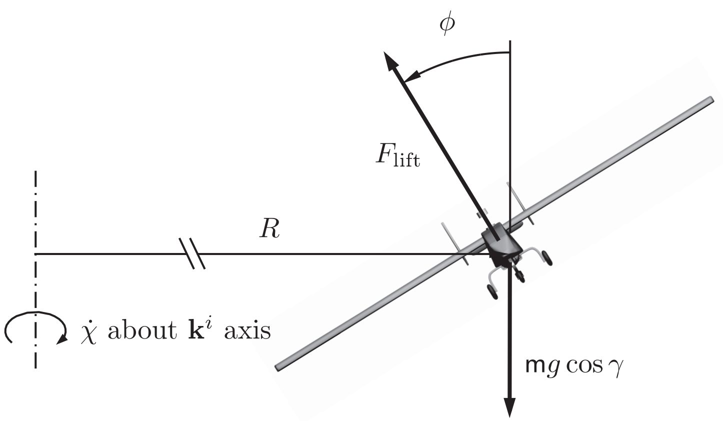
  - Sensors: rate gyros
  - Roll rate measurement

$$p_m \approx \dot{\phi} + b_p + n_p$$

  - Yaw rate measurement

$$r_m \approx \frac{g}{V} \phi + b_r + n_r$$

  - Bias terms in both measurements



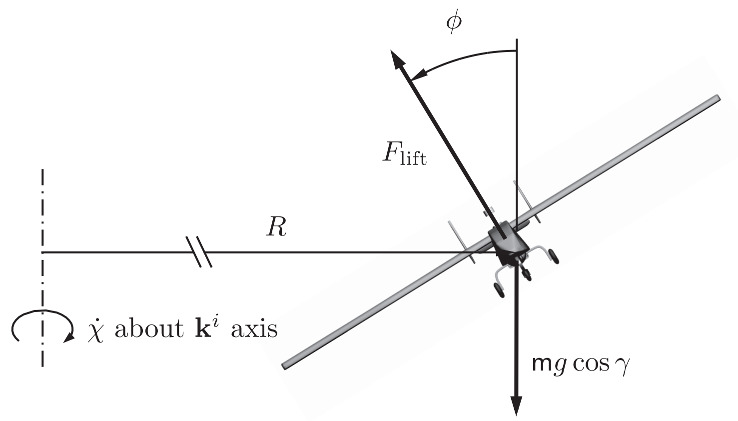
Kalman filter – exemple: roll angle measurement

- Kalman filter setup
  - Estimate roll angle and bias terms

$$\hat{\phi}(t) = ? \quad \hat{b}_p(t) = ? \quad \hat{b}_r(t) = ?$$

  - Available measurements

$$p_m = \dot{\phi} + b_p + n_p$$
$$r_m = \frac{g}{V}\phi + b_r + n_r$$



  - How does this fit the state space description for Kalman filter design?

$$\dot{x} = Ax + Bu + w \qquad E[w(t)w(\tau)^T] = \delta(t-\tau)W$$
$$y = Cx + n \qquad E[n(t)n(\tau)^T] = \delta(t-\tau)N$$
$$\dot{\hat{x}} = A\hat{x} + Bu + L(y - C\hat{x})$$

Kalman filter – exemple: roll angle measurement

- Kalman filter setup
  - Estimate roll angle and bias terms

$$\hat{\phi}(t) = ? \quad \hat{b}_p(t) = ? \quad \hat{b}_r(t) = ?$$

  - Starting point for model description

$$\dot{\phi} = p \qquad p_m = \dot{\phi} + b_p + n_p$$

$$r_m = \frac{g}{V}\phi + b_r + n_r$$

  - Identify state, input, and output for state-space description

$$\dot{x} = Ax + Bu + w$$

$$y = Cx + n$$

Try $x = \begin{bmatrix} \phi \\ b_p \\ b_r \end{bmatrix}$ $\quad y = r_m$

Kalman filter – exemple: roll angle measurement

- Kalman filter setup
  - Identify state, input, and output for state-space description

$$\dot{x} = Ax + Bu + w$$
$$y = Cx + n$$

Try $x = [\phi \; b_p \; b_r]'$ $\quad y = r_m$

  - Resulting equations

$$\dot{\phi} = p_m - b_p - n_p$$
$$y = r_m = \frac{g}{V}\phi + b_r + n_r$$
$$\dot{b}_p = n_1$$
$$\dot{b}_r = n_2$$

- Remaining questions

$$A = ? \quad B = ? \quad C = ?$$
$$u = ? \quad w \sim ? \quad n \sim ?$$

- Is the system observable?

  Check pair (A,C).

Kalman filter – exemple: roll angle measurement

- Kalman filter setup
  - Identify state, input, and output for state-space description

$$\dot{x} = Ax + Bu + w$$
$$y = Cx + n$$

Try $x = [\phi \ b_p \ b_r]'$ $\quad y = r_m$

  - Resulting equations

$$\dot{\phi} = p_m - b_p - n_p$$
$$y = r_m = \frac{g}{V}\phi + b_r + n_r$$
$$\dot{b}_p = n_1$$
$$\dot{b}_r = n_2$$

- Remaining questions

$$A = ? \quad B = ? \quad C = ?$$
$$u = ? \quad w \sim ? \quad n \sim ?$$

- Is the system observable?

No.

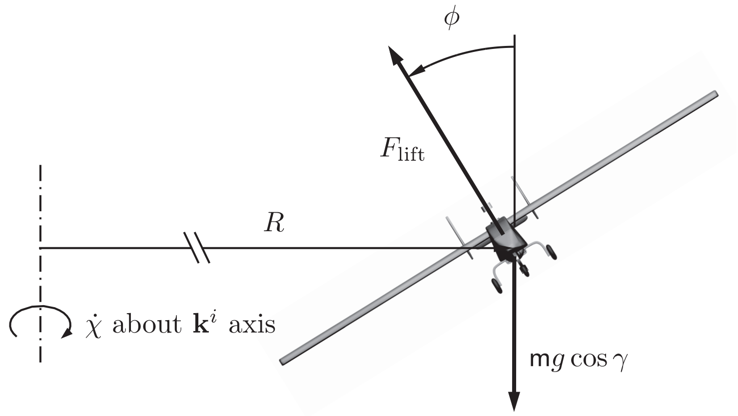
Additional sensor is needed.

Kalman filter – exemple: roll angle measurement

- Roll angle estimation for an aircraft (fixed-wing)
  - Assumption: coordinated turn, no wind, no sideslip

$$F_{lift}\cos\phi = mg$$

$$F_{lift}\sin\phi = mV^2/R = mV\dot{\psi}$$



  - Accelerometer measures

$$a_m = R^T(\ddot{p} - ge_3) + n_a$$

Neglecting $\ddot{p}$ and noise

$$a_m = -R^T ge_3 = -g\begin{bmatrix} 0 \\ \sin\phi \\ \cos\phi \end{bmatrix}$$

In fact,

$$a_m = -\frac{1}{m}\begin{bmatrix} 0 \\ 0 \\ F_{lift} \end{bmatrix} = -\begin{bmatrix} 0 \\ 0 \\ \sqrt{g^2 + (V\dot{\psi})^2} \end{bmatrix} = -g\begin{bmatrix} 0 \\ 0 \\ \sqrt{1+\tan^2\phi} \end{bmatrix}$$

Kalman filter – exemple: roll angle measurement

- Kalman filter setup (rate gyros + accelerometers)
  - Identify state, input, and output for state-space description

$$\dot{x} = Ax + Bu + w$$
$$y = Cx + n$$

Try $x = [\phi \; p \; b_p \; b_r]'$ $\quad y = [r_m \; p_m \; \phi_m]'$

  - Resulting equations

$$\begin{bmatrix} \dot{\phi} \\ \dot{p} \\ \dot{b}_p \\ \dot{b}_r \end{bmatrix} = \begin{bmatrix} p \\ w_1 \\ w_2 \\ w_3 \end{bmatrix}$$

$$y = \begin{bmatrix} \frac{g}{V}\phi + b_r + n_r \\ p + b_p + n_p \\ \phi + n_\phi \end{bmatrix}$$

- Remaining questions

$$A = ? \quad B = ? \quad C = ?$$
$$u = ? \quad w \sim ? \quad n \sim ?$$

- Is the system observable?

Yes!

Kalman filter – exemple: roll angle measurement

- Kalman filter setup (2nd alternative)
  - Identify state, input, and output for state-space description

$$\dot{x} = Ax + Bu + w$$
$$y = Cx + n$$

Try $x = [\phi \; p \; b_p \; b_r]'$ $\quad y = [r_m \; p_m \; \phi_m]'$

  - Resulting equations

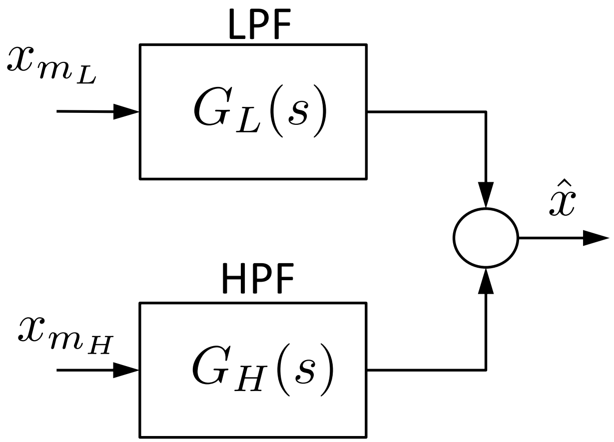
Same as before,
but using dynamic model for
time evolution of $\dot{\phi}$

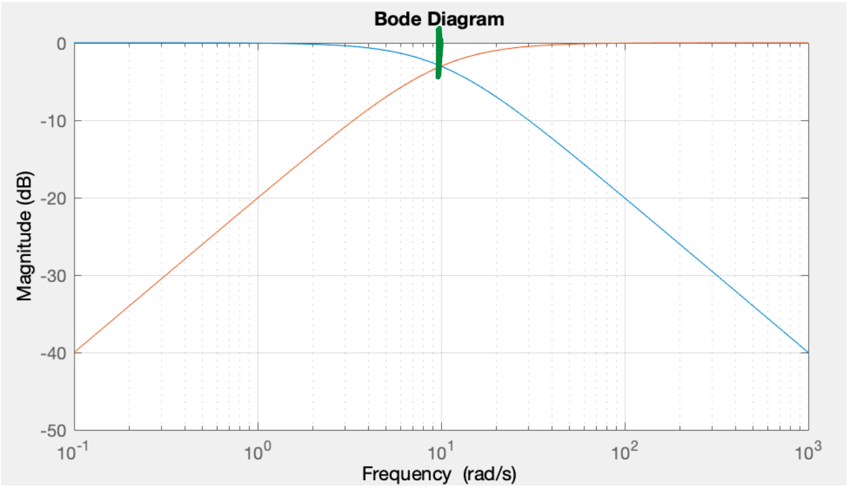
- Remaining questions

$$A = ? \quad B = ? \quad C = ?$$
$$u = ? \quad w \sim ? \quad n \sim ?$$

Complementary filters

- Context:
  - Different sensors provide reliable measurements over different (complementary) frequency ranges.
- Strategy (for a simple example):





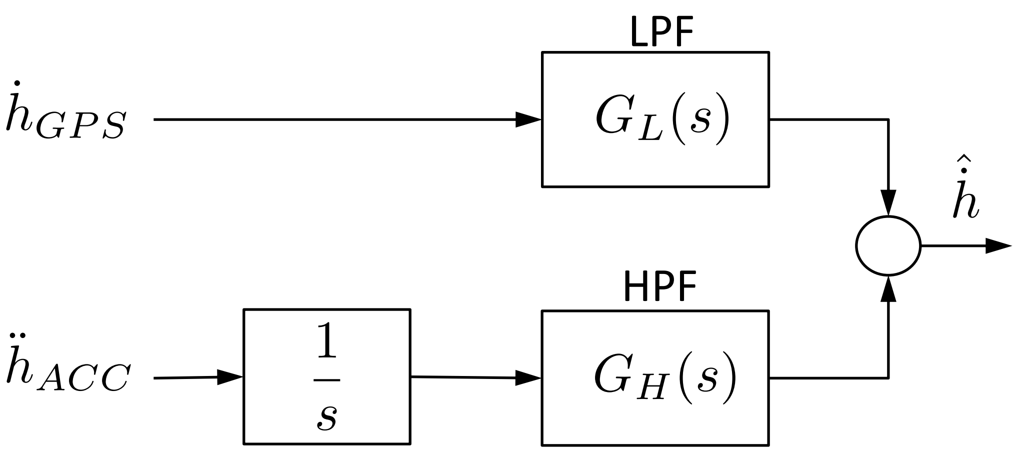
$$G_L(s) = \frac{l}{s+l} \qquad G_H(s) = \frac{s}{s+l} \qquad G_L(s) + G_H(s) = 1$$

Complementary filters

- Example: altitude rate estimation
  - low frequency from GPS receiver $\dot{h}_{GPS}$
  - high frequency integration of acceleration along the inertial z-direction

$$a_m = R(\ddot{p} - g e_3)$$

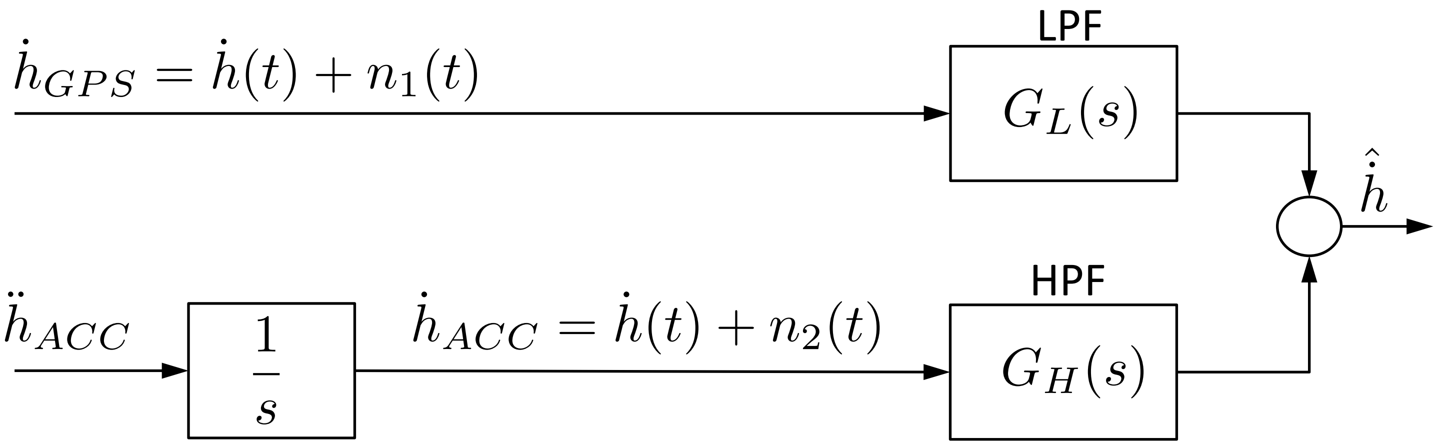
$$\ddot{h}_{ACC} = -e_3^T R_X(\hat{\phi}) R_Y(\hat{\theta}) a_m - g$$



TÉCNICO LISBOA

Complementary filters

- Example: altitude rate estimation



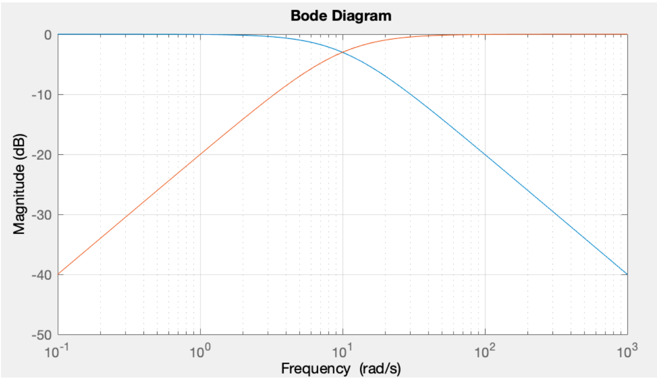
$$\hat{\dot{H}}(s) = \frac{l}{s+l}\dot{H}_1(s) + \frac{s}{s+l}\dot{H}_2(s)$$

$$= \dot{H}(s) + \frac{l}{s+l}N_1(s) + \frac{s}{s+l}N_2(s)$$

Keeps whole signal

Attenuation in the high frequencies

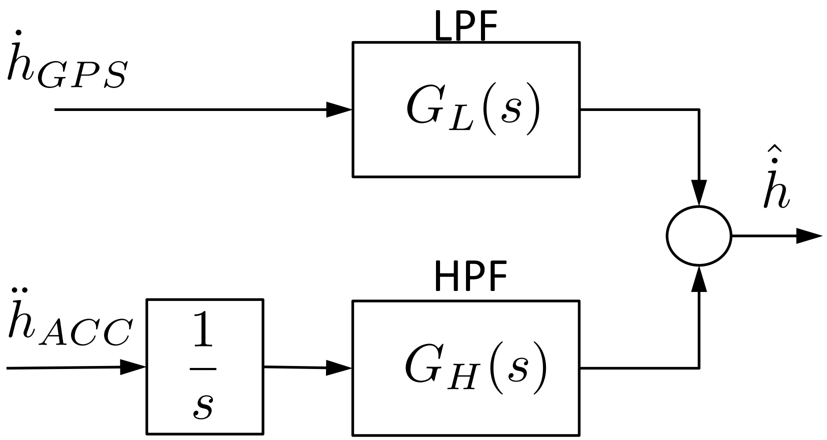
Attenuation in the low frequencies



Complementary filters and Kalman filters

- Is there a connection?

Complementary filter



Kalman filter

$$\dot{x} = Ax + Bu + w$$
$$y = Cx + n$$
$$\dot{\hat{x}} = A\hat{x} + Bu + L(y - C\hat{x})$$

with

$$x = \dot{h} \quad u = \ddot{h}_{ACC} \quad y = \dot{h}_{GPS}$$
$$\hat{x} = \hat{\dot{h}}$$

– Yes! Exercise: Compare

$$\hat{\dot{H}}(s) = \frac{l}{s+l}\dot{H}_{GPS}(s) + \frac{s}{s+l}\frac{\ddot{H}_{ACC}}{s}(s)$$

and

$$\hat{Y}(s) = \left[C(sI - A + LC)^{-1}L\right]Y(s) + \left[C(sI - A + LC)^{-1}B\right]U(s)$$

Complementary filters and Kalman filters

- It may also be applied to more evolved cases
  - State observer for roll angle and roll rate bias estimation

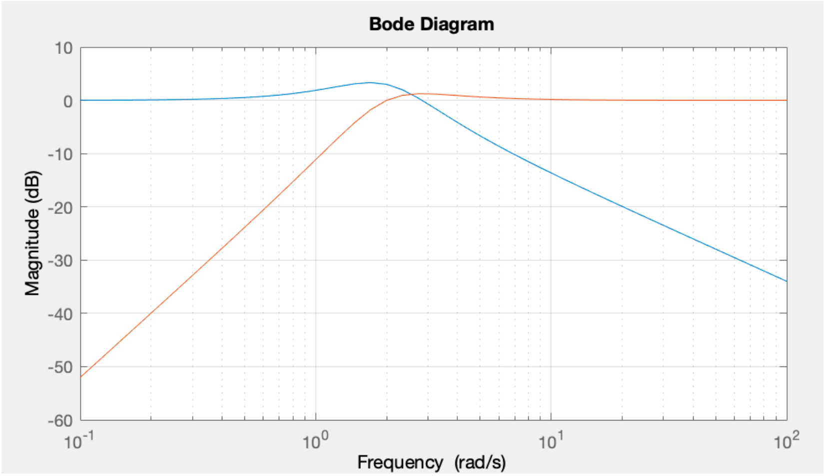
$$x = [\phi \; b_p]' \quad u = p_m \quad y = \phi_m$$

$$\hat{x} = [\hat{\phi} \; \hat{b}_p]' \quad \hat{y} = \hat{\phi}$$

$$\dot{x} = Ax + Bu + w$$

$$y = Cx + n$$

$$\dot{\hat{x}} = A\hat{x} + Bu + L(y - C\hat{x})$$



$$\hat{\Phi}(s) = \frac{l_1 s + l_2}{s^2 + l_1 s + l_2}\Phi_m(s) + \frac{s}{s^2 + l_1 s + l_2}P_m(s)$$

$$= G_L(s)\Phi_m(s) + G_H(s)\frac{P_m(s)}{s} \qquad G_L(s) + G_H(s) = 1$$